CHANNEL STRUCTURE OF ASYMPTOTIC DEFORMATIONS IN POLYNOMIAL CONTINUED FRACTION DYNAMICS: AN OUTLINE

THE SIARC AUTHOR

Abstract. We propose, define, and catalogue the notion of an *asymptotic channel* for sequences arising from polynomial continued fractions (PCFs). A channel is a triple (D, T, S) consisting of a dense subspace D of formal series, an asymptotic-gauge topology T on D, and a section S from analytic germs to D. Three concrete channels appear in the SIARC stack: the recurrence-parameter channel $L(t)$, the Borel-of-trans-series channel BoT, and the connection-coefficient channel CC. We summarise the empirical record (six $\Delta < 0$ degree-2 families surveyed in three recent computational sessions), state a conjectural no-go theorem for $L(t)$ and a speculative bridge identity between $L(t)$ and BoT, and identify the questions that the SIARC Master Conjecture's channel functor $\chi : \mathbf{\Sigma} \to \mathbf{Channel}$ must answer before the announcement paper can be written rigorously. This is an outline and program statement: no new theorems are proved.

Contents

1. Position 2
2. Definition: asymptotic channel 2
3. Catalogue of channels in the SIARC stack 4
3.1. The recurrence-parameter channel $L(t)$ 4
3.2. The Borel-of-trans-series channel BoT 4
3.3. The connection-coefficient channel CC 5
3.4. Comparison table 5
3.5. The locked WKB exponent identity 5
4. The channel functor: what we know and what we don't 6
4.1. The deformation category $\mathbf{\Sigma}$ 6
4.2. Properties of χ 6
5. Conjectured no-go theorem 7
6. Conjectured bridge identity 7
7. Related work and prior channel structures 8
7.1. Resurgence and alien calculus (Écalle, Costin, Sauzin) 9
7.2. Stokes / isomonodromic theory (Birkhoff, Jimbo–Miwa, Boalch) 9
7.3. Heun–Painlevé correspondence (Slavyanov–Lay) 9
7.4. Algebraic Painlevé moduli (Mazzocco, IKSY, Ohyama et al.) 9
8. Implications for the Master Conjecture 9
9. Open problems and program 10
Acknowledgements 11
References 11

Date: 2026-05-01 (draft).

1. Position

The SIARC stack has accumulated three empirically distinct phenomena that all advertise themselves as Painlevé reductions:

(a) The *Stokes signal* $S(t) < 1$ measured in the recurrence-parameter channel for the six known $\Delta < 0$ degree-2 polynomial continued fractions ([18], Sec. 5).
(b) The explicit reduction $V_{\text{quad}} \rightsquigarrow P_{III}(D_6)$ at the parameter point $(1/6, 0, 0, -1/2)$, established in the connection-coefficient channel via the V_{quad} Casoratian / Heun bridge ([18], Sec. 6; [26, 9]).
(c) The total failure, demonstrated in two computational sessions of 2026-05-01 ([20, 21, 22]), of both the $L(t)$ channel and a $1/n$ trans-series Borel-resummation channel to recover the V_{quad} reduction. Painlevé ansatz residuals stay flat under precision escalation (no-fit), and Padé–Borel pole radii drift unboundedly under truncation extension.

Item (c) is critical. It rules out the most naive interpretation of the Master Conjecture's *channel functor* $\chi : \mathbf{\Sigma} \to \mathbf{Channel}$ [23], namely that any deformation channel detects any reduction. The Master Conjecture asserts that the channel is a function of the deformation Σ, not of the family. The empirical record of (a)–(c) shows that χ must in fact be non-trivial: $\chi(\Sigma_{L(t)}) \neq \chi(\Sigma_{\text{CC}})$, and the two channels carry genuinely different analytic data about the same underlying sequence.

This outline catalogues what we know about the channel space, states the questions χ must answer, and isolates a small number of conjectural identities whose resolution would constitute the technical content of the announcement paper.

Scope. We restrict throughout to PCF families

$$\delta_n = b_n - \cfrac{a_n}{b_{n-1} - \cfrac{a_{n-1}}{\ddots - a_1/b_0}}$$

in the Lorentzen–Waadeland sense [12], with a_n, b_n polynomial in n over $\mathbf{Z}$, b $\mathbf{Z}$-primitive and irreducible, and (in the empirical sections) $\deg b = 2$ with discriminant $\Delta < 0$.

What is and is not in scope of this outline. This is a program statement and a definition document. We (i) fix terminology, (ii) enumerate the channels that the SIARC stack has actually built, (iii) report the empirical verdict on each, and (iv) state the open problems that the Master Conjecture's channel functor must resolve. We do *not* prove a no-go theorem (Theorem 5.1 is conjectural), we do *not* construct the bridge identity (Theorem 6.1 is speculative), and we do *not* extend the catalogue to $\deg b \geq 3$ or to families with $\Delta \geq 0$. Each of those is a self-contained follow-up listed in Section 9.

Reading guide. Section 2–Section 3 are definitional and should be read by anyone who wants to know what a SIARC "channel" is. Section 4 packages the Master Conjecture in channel language. Section 5–Section 6 are the two non-trivial conjectures whose resolution is the core technical content of the Master Conjecture announcement. Section 8–Section 9 convert the conjectures into a sequenced research program.

2. Definition: asymptotic channel

Definition 2.1 (Channel)**.** Let $X = (\delta_n)_{n \geq n_0}$ be a sequence of real or complex numbers of subexponential growth. An *asymptotic channel* for X is a triple (D, T, S) where

(1) D is a dense subspace of a fixed completion $\widehat{D}$ of formal power series (or trans-series) in a distinguished variable z, equipped with a fixed trans-series scaffold (the *template*);
(2) T is a Hausdorff topology on D induced by an asymptotic gauge $\Gamma : D \to \mathbf{R}^{\mathbf{N}}_{\geq 0}$ that compares partial sums against a chosen scale;

(3) $S : \mathcal{O}_0 \dashrightarrow D$ is a partially defined section of the natural inclusion of analytic germs at $z = 0$ into $\widehat{D}$. We write $\operatorname{dom} S \subset \mathcal{O}_0$.

The sequence X is said to *lie in the channel* (D, T, S) if there exists a germ $f \in \operatorname{dom} S$ such that $\Gamma(\widetilde{X} - S(f))$ is summable in the sense of T, where $\widetilde{X} \in \widehat{D}$ is a canonically associated formal expansion of X.

The triple (D, T, S) encodes three independent choices: *which asymptotic ansatz to fit* (D), *how to measure goodness of fit* (T), and *which analytic continuation rule to invert* (S). Different choices yield different channels even for the same underlying sequence.

Example 2.2 (Recurrence-parameter channel $L(t)$)**.** D is the space of formal series in $1/n$ with coefficients analytic in a deformation parameter $t \in (-h_0, h_0)$, with the Birkhoff–Trjitzinsky template $\delta_n(t) \sim \exp(\rho \log n) \cdot n^\sigma \cdot \sum_{k\geq 0} c_k(t)/n^k$ [2, 28]. The gauge Γ is uniform convergence on partial sums of length N, evaluated on a finite t-grid of step h. The section S is Borel–Laplace summation along $\mathbf{R}_+$ [5]. This is the channel of [18], Sec. 5.

Example 2.3 (Borel-of-trans-series channel BoT)**.** D is the space of trans-series $\log|\delta_n| \sim -An\log n + \alpha n - \beta \log n + \gamma + \sum_{k\geq 1} h_k/n^k$, with the WKB scaffold $(A, \alpha, \beta, \gamma)$ fitted first and then frozen. The gauge T measures Padé convergence of the Borel transform $\mathcal{B}(\zeta) = \sum_{k\geq 1} h_k \zeta^{k-1}/(k-1)!$, quantified by stability of the principal pole radius $|\rho_0|$ under truncation order K. The section S is Padé–Borel resummation. This is the channel of [21, 22].

Example 2.4 (Connection-coefficient channel CC)**.** Suppose δ_n is the second solution of a second-order linear difference (or differential) equation with regular singular points. D is the space of formal series in a transfer-matrix variable τ around a chosen singular point. The gauge T is convergence of Stokes constants extracted via isomonodromic deformation [10]. The section S is the Riemann–Hilbert correspondence: analytic germs at the singular point are reconstructed from the connection matrix [11, 3]. The established $V_{\text{quad}} \rightsquigarrow P_{III}(D_6)$ at $(1/6, 0, 0, -1/2)$ [9, 17] lives in this channel.

Remark 2.5 (Channel vs. summation method)*.* At first sight a channel is just a choice of summation method (Borel–Laplace vs. Padé–Borel vs. isomonodromic reconstruction). The distinction we draw is finer: two channels can use the same summation method (e.g. both $L(t)$ and BoT ultimately appeal to a Borel sum) yet target different formal-series spaces D and use different asymptotic gauges T. The empirical content of Section 1(c) is precisely that this distinction is not vacuous on the data side: $L(t)$ and BoT both fail where CC succeeds, despite all three being candidate inverses of "what δ_n is asymptotically doing".

Remark 2.6 (Why this is not a tautology)*.* The definition admits the trivial channel (D, T, S) with D full, T discrete, and S the identity on its domain, in which every sequence "lies". The non-trivial content of the channel framework is that the natural channels arising from (i) asymptotic enumeration of recurrences, (ii) Borel resummation of trans-series, and (iii) isomonodromic Stokes analysis are mutually *non-cofinal*: the partial order on channels by "inverse-image inclusion" (i.e. $(D, T, S) \leq (D', T', S')$ iff every sequence in the first channel lies in the second) does not flatten the three working channels of Section 3.1–Section 3.3 into a single one. This non-flattening is the empirical observation that motivates the bridge conjecture (Theorem 6.1) and its potential failure mode.

Remark 2.7 (Trans-series template as a choice)*.* The Birkhoff–Trjitzinsky template $\delta_n(t) \sim e^{\rho \log n} n^\sigma \sum_k c_k(t)/n^k$ is distinguished but not unique. One could equally well take a $\sqrt{n}$-template (which Session E's iteration v2 tried, before being abandoned; [21]) or a hybrid template mixing exponential and Stokes factors. Each yields a different channel by Theorem 2.1. The empirical failure of the $\sqrt{n}$-template for V_{quad} shows that channel choice is not free: a wrong template gives a well-defined but vacuous channel.

3. Catalogue of channels in the SIARC stack

We summarise the three channels in turn. Numerical values are the published values; sources are cited in line.

3.1. The recurrence-parameter channel $L(t)$.

Definition. Theorem 2.2.

Operationalisation. For each PCF family, perturb $b_n \mapsto b_n + t \cdot \omega_n$ with a chosen direction ω_n (typically a constant-term or root-radius shift), compute $\delta_n(t)$ on a finite t-grid via exact symbolic recursion, and fit a Birkhoff–Trjitzinsky tail.

Empirical findings.

- *Stokes signal.* For all six known $\Delta < 0$ degree-2 families $\{Q_{L01}, Q_{L02}, Q_{L06}, V_{\text{quad}}, Q_{L15}, Q_{L26}\}$, the Stokes proxy $S(t)$ enters the strip $S < 1$ after sufficient depth, with stability across precision sweeps [18].
- *Painlevé ansatz fit (Session D).* 0/6 families admit a Painlevé reduction at residual $\leq 10^{-4}$ under depth 3000, dps 400. V_{quad} marks marginal at 4.59×10^{-5} on a P_{III} template but does *not* recover the known $(1/6, 0, 0, -1/2)$ parameters [20].
- *Channel-typing observation.* Q_{L06} and Q_{L26} both lie in $\mathbf{Q}(\sqrt{-7})$ but disagree on the best Painlevé ansatz (P_V vs. P_{III}), so CM-field-driven Painlevé typing is ruled out at the $L(t)$ level.

Open implementation questions.

- Does enriching the deformation direction ω_n from constant-term to a moduli of perturbations (e.g. a smooth path in coefficient space) move any family into a positive $L(t)$ Painlevé fit?
- Is the $S < 1$ signal a Painlevé fingerprint at all, or is it (as Sessions D suggest) an artefact of a coarser analytic structure that does not reach the connection-coefficient resolution?

3.2. The Borel-of-trans-series channel BoT.

Definition. Theorem 2.3.

Operationalisation. Strip the $-An \log n$ factor with the closed-form WKB exponents

$$A = \begin{cases} 4 & a_n \text{ constant}, \\ 3 & a_n = c_a n + d_a, \end{cases} \qquad \alpha \;=\; A - 2\log c_b + \log |c_a|$$

verified to ≥ 13 digits across all six families [22]. Fit the residual trans-series h_k/n^k by least squares, then compute Padé approximants $[L/M]$ to the Borel transform $\mathcal{B}(\zeta)$ and probe the nearest singularity.

Empirical findings.

- *WKB exponent identity.* 6/6 families to ≥ 13 digits; this is locked into PCF-1 v1.3 and is the only positive structural result of Sessions D–E′.
- *Borel-resummation gate (V_{quad}).* Painlevé fit residuals on the Padé–Borel sum *worsen* under truncation extension ($K = 12 \to 24$): $7.2 \times 10^{-3} \to 5.0 \times 10^{-3} \to 1.0 \times 10^{-2} \to 9.9 \times 10^{-2}$. Padé pole radius diverges $2.17 \to 4.59$. Best-fit Painlevé identity flips $P_{III} \leftrightarrow P_{VI}$ [21].
- *K-scan on marginal hits.* Two $K = 12$ marginal hits ($\mathrm{Q}_{L15} \to P_V$ at 9×10^{-5}, $\mathrm{Q}_{L26} \to P_{III}$ at 3×10^{-5}) likewise drift under K-extension: residuals fail to stay $\leq 10^{-4}$, Padé pole radii drift unboundedly, ansatz identity flips [22]. Confirmed Padé-truncation artefacts.

Open implementation questions.

- Does a different summation scheme (Borel–Écalle [7, 25] rather than Padé–Borel) on the *same* trans-series space D recover Painlevé information for V_{quad}? If yes, the "channel" we tested was actually (D, T, S') with the wrong section S', not the wrong space D.

- Are the h_k stable enough across K to attempt a Borel–Laplace sum along a non-real ray? (Sessions E–E′ report $h_1 \dots h_6$ stable to 14 digits across $K = 12$–24; the channel definition is well-posed even though the downstream sum fails.)

3.3. **The connection-coefficient channel** CC.

Definition. Theorem 2.4.
Operationalisation. For families that arise as specialisations of a Heun-type linear ODE [26], compute the isomonodromic deformation parameters and read off the Painlevé moduli point.
Empirical findings.

- V_{quad}: $P_{III}(D_6)$ at $(1/6, 0, 0, -1/2)$ [9, 17].
- Other five $\Delta < 0$ degree-2 families: pipeline not yet executed (op:cc-pipeline below).

Open implementation questions.

- Build an explicit CC-channel pipeline for the five non-V_{quad} families. The pipeline requires identifying the underlying second-order linear difference equation, computing its Stokes data, and matching against the P_{II}–P_{VI} catalogue.
- Does the CC pipeline succeed on *all* of the five remaining families, on a strict subset, or on none? Each outcome implies a different shape of the channel functor χ.

3.4. **Comparison table.** The three channels can be summarised in a single table. "Detects V_{quad}'s $P_{III}(D_6)$ reduction" is the natural test question; the answer is the empirical content of Sessions D, E, E′.

Channel	Space D	Gauge T	Section S	Detects V_{quad}?
$L(t)$	series in $1/n$, t-deformed	uniform on t-grid	Borel–Laplace	no
BoT	$1/n$ trans-series of $\log\lvert\delta_n\rvert$	Padé pole-radius stability	Padé–Borel	no
CC	series in τ at sing. point	Stokes-constant convergence	Riemann–Hilbert	yes

The empirical asymmetry in the rightmost column is the single-line summary of the channel-functor problem: a rigorous χ must explain why exactly the CC row detects this reduction and the others do not.

3.5. **The locked WKB exponent identity.** A byproduct of the BoT channel construction (Section 3.2) is the explicit closed form for the WKB scaffold (A, α) extracted from $\log|\delta_n|$. We record the value because it is the only positive structural output of Sessions D–E′ and because it appears in PCF-1 v1.3 [18].

Proposition 3.1 (WKB exponent identity, empirical)**.** *For each of the six $\Delta < 0$ degree-2 families surveyed, the leading trans-series exponents of $\log|\delta_n| = -An\log n + \alpha n - \beta\log n + \gamma + \sum_{k\geq 1} h_k/n^k$ satisfy the closed form*

$$A \;=\; \begin{cases} 4 & a_n \equiv const, \\ 3 & a_n = c_a n + d_a, \end{cases} \qquad \alpha \;=\; A \;-\; 2\log c_b \;+\; \log|c_a|$$

to ≥ 13 decimal digits across the K-scan grid $K \in \{12, 16, 20, 24\}$, dps $\in \{2200, 3000, 4000, 5000\}$ [22].

Family	Δ	a_n	A	α	digits matched
Q_{L01}	-3	const	4	4	exact
Q_{L02}	-4	const	4	$4+\log 2$	13
Q_{L06}	-7	lin	3	$3-2\log 2$	14
V_{quad}	-11	const	4	$4-2\log 3$	13
Q_{L15}	-20	const	4	$4-2\log 3$	15
Q_{L26}	-28	const	4	$4+\log 3-2\log 4$	13

Theorem 3.1 is what survives of the BoT channel after the Padé–Borel resummation gate fails: the channel's *template* is well-conditioned, the h_k are stable to 14 digits across $K = 12$–24 [22], but the downstream resummation does not converge to a well-defined section S. In the language of Theorem 2.1: D and T are well-defined; S is not. This is the most refined diagnostic available for what goes wrong in BoT.

4. The channel functor: what we know and what we don't

The Master Conjecture [23] posits a functor

$$\chi : \mathbf{\Sigma} \longrightarrow \mathbf{Channel},$$

where $\mathbf{\Sigma}$ is a category of deformations of PCF families (objects: families with chosen deformation direction; morphisms: compatible reparametrisations) and **Channel** is the category whose objects are channels in the sense of Theorem 2.1.

4.1. The deformation category $\mathbf{\Sigma}$.

We make $\mathbf{\Sigma}$ slightly more precise, since the channel functor is only as well-defined as its source.

Definition 4.1 (Object of $\mathbf{\Sigma}$)**.** An object of $\mathbf{\Sigma}$ is a triple $\Sigma = (F, \omega, h)$ where $F = \mathrm{PCF}(a, b)$ is a PCF family as in the scope of Section 1, $\omega \in \mathbf{Z}[n]$ is a deformation direction, and $h > 0$ is a step parameter. The deformed family is $F_t = \mathrm{PCF}(a, b + t\omega)$ for $|t| < h$.

Definition 4.2 (Morphism of $\mathbf{\Sigma}$)**.** A morphism $(F, \omega, h) \to (F', \omega', h')$ exists when $F = F'$, $\omega' = \lambda\omega$ for $\lambda \in \mathbf{Q}^\times$, and $h' \le h/|\lambda|$. Composition is multiplication of the scalars λ. The resulting category is a Picard groupoid over each fixed family.

The channel functor χ is required to factor through the family: $\chi(F, \omega, h)$ is independent of h and invariant under positive rescalings of ω. This makes χ a functor on the quotient $\mathbf{\Sigma}/\sim$, but we retain the unquotiented form because the choice of ω class *does* affect χ (different perturbation directions can in principle produce different channels).

4.2. Properties of χ.

Conjecture 4.3 (Properties of χ)**.** *The channel functor satisfies:*

(i) Functoriality in $\mathbf{\Sigma}$. *Changing the deformation Σ on a fixed family genuinely changes the target channel $\chi(\Sigma)$. Equivalently, χ is not constant in its first argument.*

(ii) Empirical consistency. *For every PCF family F for which a Painlevé reduction is presently established (at this writing: V_{quad} alone, in the* CC *channel), χ assigns the deformation that produced that reduction to the channel in which it was established. Concretely: $\chi(\Sigma(V_{\mathrm{quad}})) = \mathrm{CC}$.*

(iii) Non-constancy. *There exist deformations Σ_1, Σ_2 with $\chi(\Sigma_1) \neq \chi(\Sigma_2)$.*

We comment on each clause.

(i) is testable: any positive $L(t)$ Painlevé fit in any new family would furnish an instance of χ taking value $L(t)$, distinguishing it from $\chi(\Sigma(V_{\mathrm{quad}})) = \mathrm{CC}$.

(ii) is currently a single data point. The cross-validation record (Section 1(c)) shows that for V_{quad} the $L(t)$ and BoT channels do not return the CC reduction; this is consistent with (ii) but does not yet test it on multiple families.

(iii) is asserted by (a)–(c) of Section 1 together with V_{quad}'s CC reduction.

What we do not know about **Channel**. The most basic open question is whether the channel space is discrete or has moduli structure. The three channels $\{L(t), \text{BoT}, \text{CC}\}$ exhausted in this outline are clearly not the only ones in principle: any choice of (D, T, S) defines a channel. Whether the *useful* set of channels (those that detect a non-trivial structure on some PCF family) is finite, countable, or has continuous moduli is open, and is the content of the (op:channel-moduli) problem in Section 9.

Worked instance of χ on the present catalogue. The minimal information required from χ on the data collected so far is the assignment

$$\chi : \big(V_{\text{quad}},\ \text{Heun-isomonodromic}\big) \longmapsto \text{CC}, \qquad \chi : \big(F_i,\ \Sigma_{\text{const-term}}\big) \longmapsto ?,$$

for $i = 1, \ldots, 6$ and the $L(t)$-style constant-term deformation $\Sigma_{\text{const-term}}$. Sessions D and E establish that the second assignment is *not* $L(t)$ (no Painlevé ansatz reaches residual $\leq 10^{-4}$ stably) and *not* BoT (Padé pole radii drift). Either χ takes value in some channel not yet in our catalogue, or the deformation $\Sigma_{\text{const-term}}$ is "channel-degenerate" in the sense that no Painlevé structure is detectable along it for these families. Theorem 5.1 below conjectures the latter for $\deg b = 2$, $\Delta < 0$; Theorem 6.1 indicates how the two readings could be reconciled via a bridge identity.

5. Conjectured no-go theorem

Conjecture 5.1 ($L(t)$ no-go for degree 2, $\Delta < 0$)**.** *Let $F = \text{PCF}(a, b)$ be a polynomial continued fraction with $a, b \in \mathbf{Z}[n]$, b irreducible and $\mathbf{Z}$-primitive, $\deg b = 2$, and discriminant $\Delta(b) < 0$. Then no Painlevé reduction of the $L(t)$-deformation $F_t = \text{PCF}(a, b + t\omega)$ exists in the $L(t)$ channel of Theorem 2.2.*

Empirical evidence. Theorem 5.1 is consistent with the Session D record [20]: 0 / 6 families admit a Painlevé fit at residual $\leq 10^{-4}$, with residuals stable under precision escalation (the diagnostic that the candidate ODE is genuinely wrong rather than under-resolved).

Falsifiability. Theorem 5.1 is falsified by a single counterexample: any $\Delta < 0$ degree-2 family for which an $L(t)$-channel Painlevé fit residual drops below 10^{-4} under precision escalation, with stable parameters.

Failure modes to monitor.

- Non-generic perturbation directions ω. Sessions B–D used D-A (constant-term) and D-B (root-radius). A generic moduli of ω has not been swept.
- Higher-rank deformations (two-parameter t). Out of scope here; would change the formal-series space D.
- Families with reducible Galois group of b. Out of scope by the irreducibility hypothesis, but worth flagging as a sanity boundary (cf. Apéry in [19]).

6. Conjectured bridge identity

Conjecture 6.1 (Bridge identity, speculative)**.** *There exists a real-analytic function $\Phi(s, \zeta_1, \ldots, \zeta_r)$ such that, for every $\Delta < 0$ degree-2 PCF family F in the regime of Theorem 5.1:*

(i) s is the limiting Stokes exponent $S(t) = \lim_{t \to 0^+} S_t$ measured in the $L(t)$ channel,
(ii) $\zeta_1, \ldots, \zeta_r$ are the singular points of the Borel transform $\mathcal{B}(\zeta)$ in the BoT *channel (if these are well-defined; presently they are not),*
(iii) $\Phi(s, \zeta_1, \ldots, \zeta_r) = 0$ holds, and
(iv) Φ-membership of F is equivalent to existence of a CC*-channel Painlevé reduction of F.*

Theorem 6.1 is the speculative deep claim. If true, the three channels $L(t)$, BoT, CC are not analytically independent: they parameterise the same underlying structure along three different probes, and the Stokes signal $S(t) < 1$ in $L(t)$ is a coarse shadow of a sharper rigidity that lives in CC.

Status. We have one data point in the affirmative direction: V_{quad} does have $S < 1$ in $L(t)$ [18] and *does* have a CC-channel $P_{III}(D_6)$ reduction [9]. The other five families have $S < 1$ but no known CC reduction yet. Theorem 6.1 predicts they all have one.

Obstructions. The main obstruction to even *stating* Theorem 6.1 precisely is that $\zeta_1, \ldots, \zeta_r$ are not well-defined for the five non-V_{quad} families: Sessions E and E′ exhibited unbounded Padé pole drift [21, 22]. Either the trans-series template is wrong, the Padé approximants are inadequate (Borel–Écalle along a complex ray would be the next test), or the BoT channel is genuinely empty for these families and the bridge identity is between $L(t)$ and CC alone, with BoT playing no role.

The connection-formula viewpoint of [4] gives the most natural formal home for Φ: Φ would be a relation among Stokes constants attached to two different singular-point neighbourhoods of the same difference equation. Whether such a relation can be written down explicitly for any $\Delta < 0$ PCF is the technical content of (op:bridge-vquad).

Three structural variants of Theorem 6.1. We distinguish three weakened forms, in increasing order of falsifiability:

(B1) *Existence.* Some real-analytic Φ as in Theorem 6.1 exists. (The default reading.)

(B2) *Algebraicity.* Φ is a polynomial in s with coefficients algebraic in $\zeta_1, \ldots, \zeta_r$. This is the strongest reasonable form and would constrain the channel functor to take values in a discrete set of algebraic strata.

(B3) *Approximate.* For some prescribed tolerance ε, $|\Phi(s, \vec{\zeta})| \leq \varepsilon$ on a dense subset of $\Delta < 0$ degree-2 families. This is the weakest form that still gives empirical traction: it would reduce χ from a deterministic functor to an ε-coarse functor and would still permit a statistical announcement.

The rubber-duck critique flagged that without committing to one of (B1)–(B3), Theorem 6.1 is too elastic to be falsifiable. We commit to (B1) as the *minimum* requirement and to (B2) as the *aspiration*; (B3) is the fallback for the announcement paper if (B1)/(B2) prove infeasible within the relay timeframe.

Sketch of Φ for V_{quad}. For V_{quad} the CC-channel reduction parameters $(1/6, 0, 0, -1/2)$ in $P_{III}(D_6)$ are exactly determined (via the Heun pathway of [26] and the IKSY classification [9, 17]). From Theorem 3.1, $A = 4$ and $\alpha = 4 - 2\log 3$. The $L(t)$ Stokes proxy $S(t)$ measured in [18] satisfies $S < 1$ unconditionally. A candidate skeleton for Φ would therefore relate S, A, α, and a (currently undefined) Borel singular point ζ_0 via

$$\Phi_V(s, \zeta_0) \stackrel{?}{=} s + A^{-1}\big(\alpha - 2\log c_b\big) - \Re(\zeta_0)/A.$$

We stress that this is a structural placeholder, not a claim: the right-hand side is gauge-fixed to vanish for V_{quad} by construction, and its empirical content lives in whether the same shape with A, α replaced by the row values of Theorem 3.1 produces a ζ_0 that actually appears as a Borel singularity for the other five families. As of Session E′, no such ζ_0 exists at the Padé level, so the sketch is currently untestable.

7. Related work and prior channel structures

The channel framework of Theorem 2.1 did not arise in a vacuum. We sketch four lines of prior work that supply either the formal vocabulary or a non-PCF analogue, and we flag where the SIARC channels diverge from each.

7.1. **Resurgence and alien calculus (Écalle, Costin, Sauzin).** The most developed pre-existing analogue is the resurgence-theoretic framework of [7, 5, 25, 16]. There a divergent series is identified with its Borel transform, analytic continuation along admissible rays defines a collection of summed functions, and the differences between those sums (Stokes constants) carry the deep arithmetic content. Our BoT channel is essentially a discrete Padé-truncated version of this construction; the failure mode observed in Session E ([21]) is consistent with the well-known phenomenon that Padé approximation does not always recover the alien structure even when the underlying series is resurgent. The continuous-side success of resurgence [6] on P_I is the structural benchmark to which any positive resolution of (op:bridge-vquad) must aspire.

7.2. **Stokes / isomonodromic theory (Birkhoff, Jimbo–Miwa, Boalch).** The connection-coefficient channel CC has a long history in the theory of isomonodromic deformations of linear ODEs [10, 27]. Boalch's work [3] packages the Stokes data as a Poisson–Lie structure on a moduli space, which is the natural ambient candidate for the channel-moduli question (op:channel-moduli). The non-classical aspect for PCFs is that the underlying object is a second-order linear *difference* (not differential) equation; the isomonodromic theory transfers but is not entirely streamlined.

7.3. **Heun–Painlevé correspondence (Slavyanov–Lay).** [26] treats Heun-class linear ODEs with four regular singular points and their reductions to the Painlevé family. The V_{quad} Casoratian arises as a specialisation of this framework, and that specialisation is how the $(1/6, 0, 0, -1/2)$ moduli point is computed in practice. Whether the other five $\Delta < 0$ degree-2 PCFs admit Heun-style specialisations of any kind is an empirical question that (op:cc-pipeline) would answer.

7.4. **Algebraic Painlevé moduli (Mazzocco, IKSY, Ohyama et al.)** [13, 14, 9, 17, 8] give finite catalogues of algebraic and Picard-type solutions to the Painlevé equations together with the Stokes-data parameters that realise them. Our T2B / UMB-PVI-MATCH umbrella result [24] is an attempt to read PCF families *into* this catalogue: Class B (16/16) lands inside $P_{III}(D_6)$, and a single P_{VI}-Picard rigidity candidate (A052) was flagged for further screening. That candidate, if it survives, will be the second example beyond V_{quad} where χ takes value in CC on a specific PCF.

The novel content of the present outline is not in the formal-series machinery, which is well established, but in the *empirical* observation that on the natural $\Delta < 0$ degree-2 PCF stratum the three classical channels are *empirically separated* on the V_{quad}-detection task. Whether this separation is intrinsic to PCFs or an artefact of the channels we have chosen to test is the central open question.

8. Implications for the Master Conjecture

A rigorous announcement of the Master Conjecture [23] requires the following intermediate results, in increasing order of difficulty:

(1) **Pipeline closure.** An explicit CC-channel pipeline for the five non-V_{quad} degree-2 $\Delta < 0$ families (op:cc-pipeline). This is the *minimum* test of χ: Theorem 4.3(ii) demands that whatever the pipeline returns is the value of χ on those families' deformations.
(2) **No-go proof or refutation.** Theorem 5.1 either proved (settling that $L(t)$ never detects Painlevé in this regime) or refuted by a positive counterexample (op:no-go-proof). Either outcome sharpens Theorem 4.3(i).
(3) **Partial bridge identity.** A version of Theorem 6.1 for at least one family beyond V_{quad} (op:bridge-vquad). This is the first non-trivial test of the speculative bridge.
(4) **Cubic extension.** Channel catalogue extended to $\deg b \geq 3$ along the lines of [19] (op:cubic-channels).

In the language of Section 4: items 1 and 2 discharge Theorem 4.3(i)–(ii) for one stratum ($\deg = 2$, $\Delta < 0$); item 3 starts to characterise the target category **Channel**; item 4 begins to extend χ across the natural product structure of $\mathbf{\Sigma}$.

The announcement paper cannot be written rigorously without *at least* item 1 above (the CC pipeline), because otherwise the empirical content of Theorem 4.3(ii) collapses to a single example.

9. Open problems and program

We collect six concrete open problems with tractability estimates. "Session" here means one SIARC relay session (of order 10 hours of agent compute plus human-side review).

Open problem 9.1 (op:no-go-proof)**.** Prove or refute Theorem 5.1. *Tractability.* A refutation may be within reach in 1–3 sessions if any moduli of perturbation directions ω produces a stable $L(t)$ Painlevé fit. A proof is a multi-year program: it requires structural understanding of why the Birkhoff–Trjitzinsky template's gauge cannot distinguish Painlevé equations within the $\Delta < 0$ regime.

Open problem 9.2 (op:bridge-vquad)**.** Construct a candidate Φ in Theorem 6.1 for V_{quad} alone, using its known CC reduction as the target. *Tractability.* 2–5 sessions for a candidate, if a suitable Borel–Écalle resummation can be implemented along a complex ray; multi-year for a proof.

Open problem 9.3 (op:cc-pipeline)**.** Implement an explicit CC-channel pipeline for the five non-V_{quad} degree-2 $\Delta < 0$ families. Report which (if any) admit a Painlevé reduction. *Tractability.* 1–2 sessions per family for the straightforward Heun-type cases (cf. V_quad's [26] pathway); harder if the underlying linear difference equation does not arise from classical Heun specialisation.

Open problem 9.4 (op:channel-moduli)**.** Is the channel space **Channel** a moduli space (with positive-dimensional families of channels) or a discrete / finite set? Equivalently, does varying (D, T, S) continuously through admissible triples give continuous deformations of the channel? *Tractability.* Multi-year. Likely requires the Stokes-matrix / Frobenius-manifold framework of [3] as a guide, and a working CC pipeline as an empirical anchor.

Open problem 9.5 (op:cubic-channels)**.** Do PCF-2 cubic families ([19]) admit the same three-channel structure $\{L(t), \mathrm{BoT}, \mathrm{CC}\}$, or does $\deg b = 3$ introduce qualitatively new channels? *Tractability.* 3–5 sessions for a first scan once the PCF-2 enumeration cubic Apery-positive-control pipeline is in place.

Open problem 9.6 (op:partition-link, speculative)**.** Does the Meinardus partition-asymptotic universality of G-01 [15, 1] live in any of the channels of Section 3? If so, the channel functor χ extends across SIARC Path 5 and unifies the partition and PCF strands. *Tractability.* Multi-year. Requires reformulating Meinardus universality as a channel-detected property of a suitable trans-series.

Open problem 9.7 (op:de-borel)**.** Implement Borel–Écalle ray summation [7, 25] as a *distinct* section S' on the same trans-series template D as BoT. Test on V_{quad}. If S' recovers the CC moduli point, the SIARC catalogue gains a fourth channel $\mathrm{BoT}' = (D, T', S')$ and the empirical asymmetry of Section 3.4 sharpens to a statement about Padé gauges specifically, not about Borel resummation in general. *Tractability.* 2–4 sessions; well within relay scope, and likely the highest-value next experiment.

Open problem 9.8 (op:zero-one-law)**.** Is there a zero-one law for channel detection on a fixed PCF family? Concretely: given a family F, is the set of deformations Σ for which $\chi(\Sigma) \neq \mathsf{trivial}$ either of measure zero or of full measure in a natural sense? A positive answer would force χ to factor through a discrete invariant of the family. *Tractability.* Multi-year and currently lacks a candidate measure on $\mathbf{\Sigma}$.

Acknowledgements

We thank the SIARC relay infrastructure (papanokechi/siarc-relay-bridge) for storing and checkpointing the empirical record cited throughout.

AI disclosure. This outline was drafted with the assistance of large language models in the SIARC self-iterating analytic relay chain. All numerical claims trace to exact computations stored under `sessions/2026-05-01/` on the SIARC bridge. The present document contains no new theorems and no new numerical results; it organises and frames existing empirical and structural material into a program statement.

References

[1] George E. Andrews. *The theory of partitions*. Addison-Wesley, 1976.
[2] George D. Birkhoff and Waldemar J. Trjitzinsky. Analytic theory of singular difference equations. *Acta Math.*, 60:1–89, 1932.
[3] Philip Boalch. Stokes matrices, Poisson Lie groups and Frobenius manifolds. *Invent. Math.*, 146(3):479–506, 2001.
[4] O. Costin and R. D. Costin. On the formation of singularities of solutions of nonlinear differential systems in antistokes directions. *Invent. Math.*, 145(3):425–485, 2001.
[5] Ovidiu Costin. *Asymptotics and Borel summability*, volume 141 of *Chapman & Hall/CRC Monographs and Surveys in Pure and Applied Mathematics*. CRC Press, Boca Raton, FL, 2008.
[6] Éric Delabaere. *Divergent series, summability and resurgence III: Resurgent methods and the first Painlevé equation*, volume 2155 of *Lecture Notes in Mathematics*. Springer, 2016.
[7] Jean Écalle. Les fonctions résurgentes. *Publ. Math. Orsay*, 81-05, 81-06, 85-05, 1981–1985.
[8] Davide Guzzetti. A review of the sixth Painlevé equation. *Constr. Approx.*, 41(3):495–527, 2015.
[9] Katsunori Iwasaki, Hironobu Kimura, Shun Shimomura, and Masaaki Yoshida. *From Gauss to Painlevé: A modern theory of special functions*. Vieweg, Braunschweig, 1991.
[10] Michio Jimbo and Tetsuji Miwa. Monodromy preserving deformation of linear ordinary differential equations with rational coefficients. II. *Phys. D*, 2(3):407–448, 1981.
[11] Michèle Loday-Richaud. *Divergent series, summability and resurgence II: Simple and multiple summability*, volume 2154 of *Lecture Notes in Mathematics*. Springer, 2016.
[12] Lisa Lorentzen and Haakon Waadeland. *Continued fractions, Vol. 1: Convergence theory*. Atlantis Press, 2nd edition, 2008.
[13] Marta Mazzocco. Picard and Chazy solutions of Painlevé VI. *Math. Ann.*, 321(1):157–195, 2001.
[14] Marta Mazzocco. Painlevé sixth equation as isomonodromic deformations equation of an irregular system. *New Trends in Integrability and Partial Solvability (NATO Sci. Ser. II)*, 132:219–238, 2004.
[15] Günter Meinardus. Asymptotische Aussagen über Partitionen. *Math. Z.*, 59:388–398, 1954.
[16] Claude Mitschi and David Sauzin. *Divergent series, summability and resurgence I: Monodromy and resurgence*, volume 2153 of *Lecture Notes in Mathematics*. Springer, 2016.
[17] Yousuke Ohyama, Hiroyuki Kawamuko, Hidetaka Sakai, and Kazuo Okamoto. Studies on the Painlevé equations V: third Painlevé equations of type $D_7^{(1)}$ and $D_8^{(1)}$. *J. Math. Sci. Univ. Tokyo*, 13:145–204, 2006.
[18] Papá-Nokechi. PCF-1 v1.3: Stokes signal and channel scope for $\delta < 0$ degree-2 polynomial continued fractions. Zenodo working draft (DOI pending), 2026.
[19] Papá-Nokechi. PCF-2 program statement: cubic polynomial continued fractions and the Galois stratification conjecture, 2026.
[20] Papá-Nokechi. Session D: Painlevé-deep 6× across $\delta < 0$ degree-2 PCFs (L(t) channel). SIARC bridge, sessions/2026-05-01/PAINLEVE-DEEP-6X/, 2026.
[21] Papá-Nokechi. Session E: Borel channel 5× V_quad cross-validation. SIARC bridge, sessions/2026-05-01/BOREL-CHANNEL-5X/, 2026.
[22] Papá-Nokechi. Session E′: Borel channel K-scan and Padé-truncation diagnosis. SIARC bridge, sessions/2026-05-01/BOREL-CHANNEL-K-SCAN/, 2026.
[23] Papá-Nokechi. SIARC umbrella: a master conjecture on functorial channel assignment in asymptotic deformation theory of polynomial continued fractions, 2026.
[24] Papá-Nokechi. T2B v3: Class-B saturation and the P-06 / 140-family Painlevé fingerprint, 2026.
[25] David Sauzin. Resurgent functions and splitting problems. In *RIMS Kôkyûroku Bessatsu*, volume B40, pages 197–250. 2014.
[26] S. Yu. Slavyanov and Wolfgang Lay. *Special functions: a unified theory based on singularities*. Oxford University Press, Oxford, 2000.

[27] Marius van der Put and Michael F. Singer. Galois theory of linear differential equations. *Grundlehren der mathematischen Wissenschaften*, 328, 2003.
[28] Jet Wimp. *Computation with recurrence relations*. Pitman, Boston, 1984.